

Image Cost Budget — the ~\$10-15 One-Time Posture

This is the exact configuration to generate **all** catalog images once, for the whole 88-country catalog, for about **\$10-15 total** (one time). It is now the **default** in the code — you only need to add your AWS credentials to activate it. If you don't add AWS creds, the catalog simply launches **text-only at \$0** and you can turn images on later.

What gets generated (and why it's cheap)

- **~905 product images** — one original image per product. Thanks to the *template-once* design, that single image is reused across all 88 countries (only the flag/price change), so it is **not** multiplied by country.
- **~40 top-level department tile images** — subcategory tiles are OFF by default in this posture.
- **0 images for the ~21,700 browse nodes** — those are text, not images.
- **Total ≈ 945 images × ~\$0.01 (Titan Image) ≈ \$9-10 one-time** (headroom to ~\$15).

If you later flip subcategory tiles on, add ~905 more images (~\$9 more) for ~1,850 total (~\$18).

The exact settings (already the defaults)

Set these in the Railway Variables UI (or backend/.env). The first two plus AWS creds are all that's strictly required — the rest already default to these values.

```
IMAGE_PROVIDER=aws_bedrock
IMAGE_MODEL=amazon.titan-image-generator-v1
CATALOG_IMAGES_ENABLED=1           # 0 = text-only launch at $0
CATALOG_IMAGES_MAX_PER_RUN=100     # paces cost across scheduled runs
CATALOG_SUBCATEGORY_IMAGES=0       # top-level tiles only (cheap). 1
= also ~905 subcategory tiles
CATALOG_IMAGE_SIZE=1024x1024

# AWS credentials that activate Bedrock (an IAM user with
bedrock:InvokeModel):
AWS_REGION=us-east-1               # a region where Titan Image is
enabled
AWS_ACCESS_KEY_ID=...
AWS_SECRET_ACCESS_KEY=...
```

Also in the AWS console: open **Amazon Bedrock → Model access** and enable **Amazon Titan Image Generator** in that region. Optional: set `S3_BUCKET` + `S3_PUBLIC_BASE` so images are stored as short URLs instead of inline.

These are also editable live in **Admin → Platform Settings** (AI & Agents section) without a redeploy.

How the cost stays bounded

- **Template-once:** an image is generated a single time and reused across every country — not per country (per-country would be

~\$3,200; this posture is ~\$10).

- **CATALOG_IMAGES_MAX_PER_RUN=100:** at most 100 images per scheduled run, so the ~945 fill over ~10 runs — no single spike.
- **CATALOG_SUBCATEGORY_IMAGES=0:** skips the ~905 subcategory tiles you don't need at launch.
- **One-time:** once generated, images are reused forever; ongoing cost is only for new products you add.
- **Kill switch:** set CATALOG_IMAGES_ENABLED=0 to launch text-only at \$0 and backfill later.

Cheaper / premium alternatives

- **Even cheaper:** self-host SDXL on a scale-to-zero SageMaker endpoint (IMAGE_PROVIDER=aws_sagemaker
◦ SAGEMAKER_IMAGE_ENDPOINT) — roughly \$0.002/image, so all images ≈ \$2-8 plus a little warm time.
- **Premium look:** IMAGE_MODEL=amazon.nova-canvas-v1:0 (~\$0.04/image) → all images ≈ \$40-75.

AWS list prices change; confirm the current Bedrock image pricing before a large run. The image *count* above comes directly from this app's catalog settings.